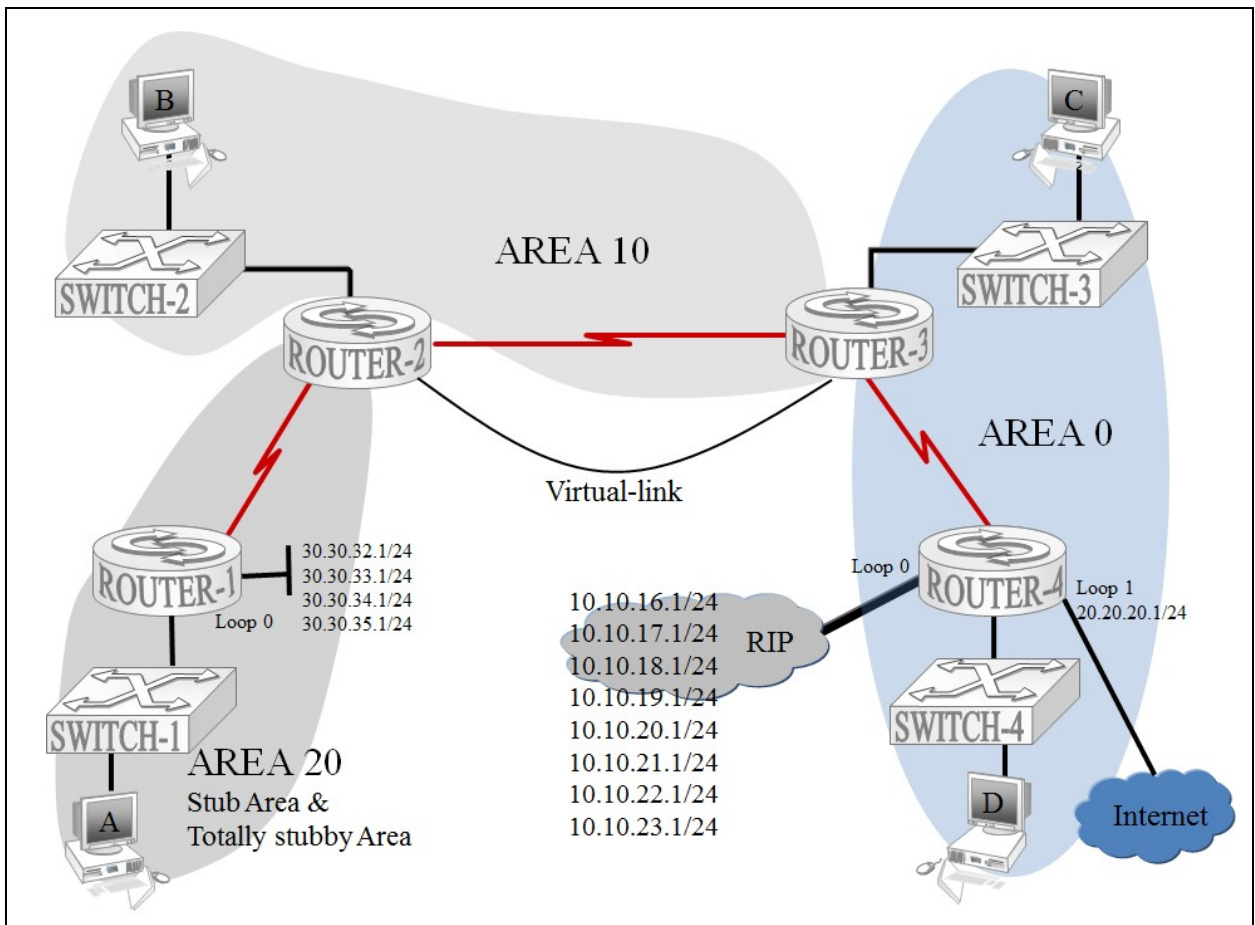
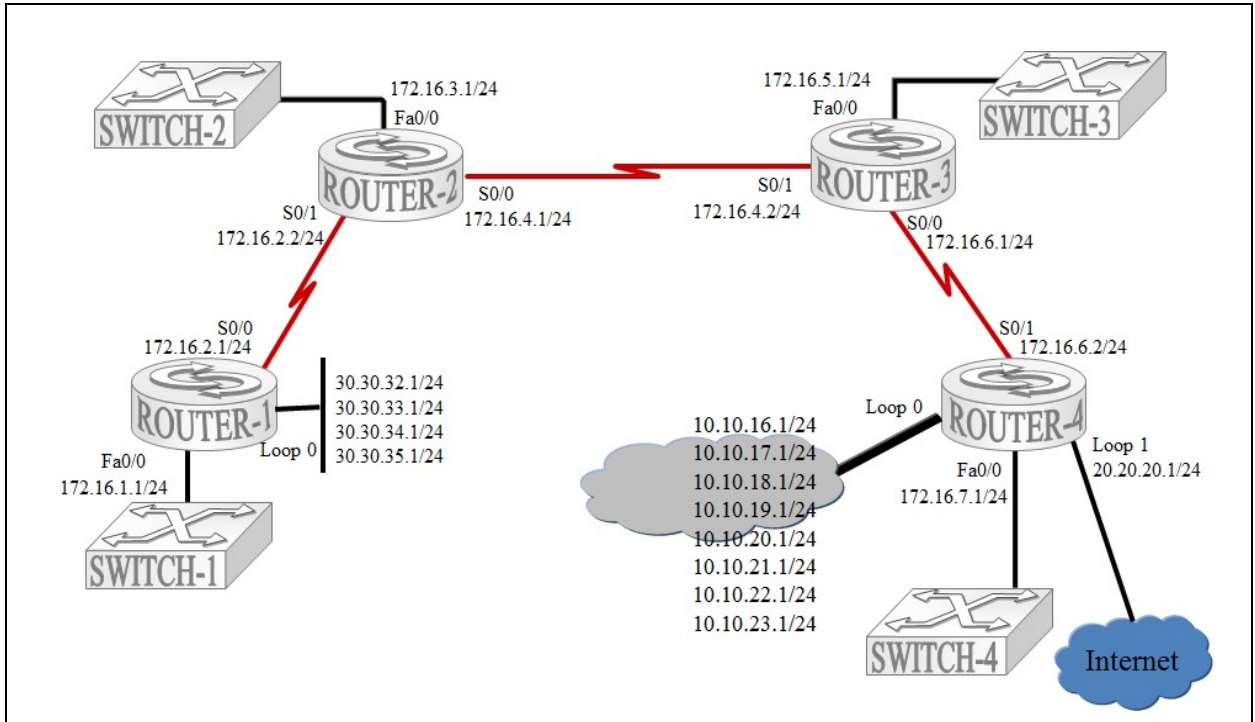


Multi-Area OSPF



❶ [R1/R2/R3/R4] 인터페이스 설정

R1	R2
<pre>#conf t (config)#int loop 0 (config-if)# ip address 30.30.32.1 255.255.255.0 (config-if)# ip address 30.30.33.1 255.255.255.0 sec (config-if)#ip address 30.30.34.1 255.255.255.0 sec (config-if)#ip address 30.30.35.1 255.255.255.0 sec (config-if)# int fa0/0 (config-if)# ip address 172.16.1.1 255.255.255.0 (config-if)# no shut (config-if)# int s0/0 (config-if)#ip address 172.16.2.1 255.255.255.0 (config-if)# no shut (config-if)#end #copy run start</pre>	<pre>#conf t (config)#int fa0/0 (config-if)#ip address 172.16.3.1 255.255.255.0 (config-if)#no shut (config-if)# int s0/0 (config-if)#ip address 172.16.4.1 255.255.255.0 (config-if)#no shut (config-if)# int s0/1 (config-if)# ip address 172.16.2.2 255.255.255.0 (config-if)# no shut (config-if)#end #copy run start</pre>

R3	R4
<pre>#conf t (config)#int fa0/0 (config-if)# ip address 172.16.5.1 255.255.255.0 (config-if)# no shut (config-if)# int s0/0 (config-if)# ip address 172.16.6.1 255.255.255.0 (config-if)# no shut (config-if)# int s0/1 (config-if)# ip address 172.16.4.2 255.255.255.0 (config-if)# no shut (config-if)#end #copy run start</pre>	<pre>#conf t (config)#int loop 0 (config-if)#ip address 10.10.16.1 255.255.255.0 (config-if)#ip address 10.10.17.1 255.255.255.0 sec (config-if)#ip address 10.10.18.1 255.255.255.0 sec (config-if)#ip address 10.10.19.1 255.255.255.0 sec (config-if)#ip address 10.10.20.1 255.255.255.0 sec (config-if)#ip address 10.10.21.1 255.255.255.0 sec (config-if)#ip address 10.10.22.1 255.255.255.0 sec (config-if)#ip address 10.10.23.1 255.255.255.0 sec (config-if)#int loop 1 (config-if)#ip address 20.20.20.1 255.255.255.0 (config-if)#no shut (config-if)#int fa0/0 (config-if)#ip address 172.16.7.1 255.255.255.0 (config-if)#no shut (config-if)#int s0/1 (config-if)#ip address 172.16.6.2 255.255.255.0 (config-if)#no shut (config-if)#end #copy run start</pre>

② [R1/R2/R3/R4] Multi- Area OSPF & Virtual link

R1	<pre>#conf t (config)#router ospf 1 (config-router)#network 30.30.32.0 0.0.3.255 area 20 (config-router)#network 172.16.1.0 0.0.0.255 area 20 (config-router)#network 172.16.2.0 0.0.0.255 area 20 #end #copy run start</pre>
R2	<pre>#conf t (config)#router ospf 2 (config-router)#router-id 172.16.3.1 (config-router)#area 10 virtual-link 172.16.5.1 (config-router)#network 172.16.2.0 0.0.0.255 area 20 (config-router)#network 172.16.3.0 0.0.0.255 area 10 (config-router)#network 172.16.4.0 0.0.0.255 area 10 #end #copy run start</pre>
R3	<pre>#conf t (config)#router ospf 3 (config-router)#router-id 172.16.5.1 (config-router)#area 10 virtual-link 172.16.3.1 (config-router)#network 172.16.4.0 0.0.0.255 area 10 (config-router)#network 172.16.5.0 0.0.0.255 area 0 (config-router)#network 172.16.6.0 0.0.0.255 area 0 #end #copy run start</pre>
R4	<pre>#conf t (config)#router ospf 4 (config-router)#network 172.16.6.0 0.0.0.255 area 0 (config-router)#network 172.16.7.0 0.0.0.255 area 0 #end #copy run start</pre>

③ [R4] RIP & Redistribute

R4	<pre>#conf t (config)#router rip (config-router)#network 10.0.0.0 (config-router)#router ospf 4 (config-router)#redistribute rip subnets #end #copy run start</pre>
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④ [R4] Default route

R4	<pre>#conf t (config)#ip route 0.0.0.0 0.0.0.0 20.20.20.2 (config)#router ospf 4 (config-router)#default-information originate (config-router)#end #end #copy run start</pre>
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⑤ [R2/R4] Route Summarization

R2	<pre>#conf t (config)#router ospf 2 (config-router)#area 20 range 30.30.32.0 255.255.252.0 (config-router)#end #end #copy run start</pre>	ABR 요약
R4	<pre>#conf t (config)#router ospf 4 (config-router)#summary-address 10.10.16.0 255.255.248.0 (config-router)#end #end #copy run start</pre>	ASBR 요약

⑥ Area 20 Stub area

R1	R2
<pre>#conf t (config)#router ospf 1 (config-router)#area 20 stub (config-router)#end #end #copy run start</pre>	<pre>#conf t (config)#router ospf 2 (config-router)#area 20 stub (config-router)#end #end #copy run start</pre>

⑦ Area 20 Totally stubby area

R1	R2
<pre>#conf t (config)#router ospf 1 (config-router)#area 20 stub (config-router)#end #end #copy run start</pre>	<pre>#conf t (config)#router ospf 2 (config-router)#area 20 stub no-summary (config-router)#end #end #copy run start</pre>